

SQL Scenario-Based Assessment (DDL, DML, DQL, DCL & TCL)

Scenario 1 - Employee Management System

Scenario

You have joined an IT company as a Database Developer.

The HR team wants to maintain employee information.

Create an **Employee** table and insert the given records.

Later, HR asks you to display all employee details.

Table to Create

Employee

Column	Data Type
EmpID	INT
EmpName	VARCHAR(50)
Department	VARCHAR(30)
Salary	INT

Records to Insert

EmpID	EmpName	Department	Salary
101	John	IT	45000
102	Mary	HR	35000
103	David	Finance	55000
104	Sam	IT	40000

EmpID	EmpName	Department	Salary
105	Priya	HR	38000

Question

Display all employee records.

SQL Command Type: DQL

Scenario 2 - College Student Database

Scenario

A college wants to maintain student details.

Create a **Student** table and insert the following records.

Later, the college decides to add an Email column.

Table

Student

Column	Data Type
StudentID	INT
StudentName	VARCHAR(50)
Course	VARCHAR(30)

Records

StudentID	StudentName	Course
1	Rahul	SQL
2	Priya	Python
3	Arun	Power BI

StudentID	StudentName	Course
4	Sneha	Java
5	Karthik	SQL

Question

Modify the Student table by adding a new column called **Email**.

SQL Command Type: DDL

Scenario 3 - Online Shopping System

Scenario

An online shopping company stores product information.

Create a **Product** table and insert the following records.

Later, the manager wants to increase the price of Product ID 103.

Table

Product

Column	Data Type
ProductID	INT
ProductName	VARCHAR(50)
Price	INT

Records

ProductID	ProductName	Price
101	Laptop	60000
102	Mouse	800

ProductID	ProductName	Price
103	Keyboard	1200
104	Monitor	15000
105	Printer	9000

Question

Update the price of Product ID 103 to **1500**.

SQL Command Type: DML

Scenario 4 - Hospital Database

Scenario

A hospital stores patient details.

Create a **Patient** table and insert the following records.

Later, Patient ID 2 requests that their record be removed from the system.

Table

Patient

Column	Data Type
PatientID	INT
PatientName	VARCHAR(50)
Disease	VARCHAR(50)

Records

PatientID	PatientName	Disease
1	Ramesh	Fever

PatientID	PatientName	Disease
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2	Suresh	Cold
3	Anitha	Diabetes
4	Meena	Asthma
5	Kumar	Typhoid

Question

Delete Patient ID 2 from the table.

SQL Command Type: DML

Scenario 5 - School Management System

Scenario

A school created a temporary table for annual event registration.

After the event, the school wants to remove all records but keep the table for next year.

Table

EventRegistration

Column	Data Type
StudentID	INT
StudentName	VARCHAR(50)
EventName	VARCHAR(50)

Records

StudentID	StudentName	EventName
1	Rahul	Dance

StudentID	StudentName	EventName
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2	Priya	Singing
3	Arun	Drawing
4	Sneha	Quiz
5	Karthik	Drama

Question

Remove all records from the table without deleting the table structure.

SQL Command Type: DDL

Scenario 6 - Company Database

Scenario

The company wants to rename its existing Employee table to Staff.

Table

Employee

EmpID	EmpName	Salary
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101	John	45000
102	Mary	35000
103	David	50000
104	Sam	42000
105	Priya	39000

Question

Rename the Employee table to **Staff**.

SQL Command Type: DDL

Scenario 7 - Banking System

Scenario

A bank manager updates a customer's account balance.

The manager wants the changes to be permanently saved in the database.

Table

Account

Column	Data Type
AccountNo	INT
CustomerName	VARCHAR(50)
Balance	INT

Records

AccountNo	CustomerName	Balance
1001	John	50000
1002	Mary	30000
1003	David	70000
1004	Sam	45000
1005	Priya	60000

Question

Update Account No. 1002's balance to **35000** and permanently save the changes.

SQL Command Type: TCL

Scenario 8 - Library Management System

Scenario

A librarian accidentally deletes one book record.

Before committing the transaction, they realize the mistake and want to restore the deleted record.

Table

Book

Column	Data Type
BookID	INT
BookName	VARCHAR(50)
Author	VARCHAR(50)

Records

BookID	BookName	Author
1	SQL Basics	James
2	Python Guide	Robert
3	Java Programming	John
4	Power BI	David
5	Data Science	Peter

Question

Delete Book ID 3 and restore it before committing the transaction.

SQL Command Type: TCL

Scenario 9 - Company Security

Scenario

A new employee named **Student1** has joined the reporting team.

The manager wants the employee to view employee details but should not modify any records.

Existing Table

Employee

Sample Records

EmpID	EmpName	Department	Salary
101	John	IT	45000
102	Mary	HR	35000
103	David	Finance	50000
104	Sam	IT	42000
105	Priya	HR	38000

Question

Give **Student1** permission to view the Employee table only.

SQL Command Type: DCL

Scenario 10 - E-Commerce Database

Scenario

A junior developer accidentally creates a table called **CustomerDetails2026**.

The company decides to delete the entire table because it is no longer required.

Table

CustomerDetails2026

CustomerID	CustomerName	City
1	John	Chennai
2	Mary	Bangalore
3	David	Hyderabad
4	Sam	Coimbatore
5	Priya	Madurai

Question

Delete the entire table permanently from the database.

SQL Command Type: DDL

Command Coverage

Scenario SQL Command Type

- 1 DQL (SELECT)
- 2 DDL (ALTER)
- 3 DML (UPDATE)
- 4 DML (DELETE)
- 5 DDL (TRUNCATE)
- 6 DDL (RENAME)
- 7 TCL (COMMIT)
- 8 TCL (ROLLBACK)
- 9 DCL (GRANT)

Scenario SQL Command Type

10 DDL (DROP)